



FAMILY FACT FILE

Key features

- Mouths but no jaws
- Many species lacked paired fins
- Usually no stomach
- Swam by beating a muscular tail

When

Some jawless fish fossils have been dated to the Cambrian Period, more than 500 million years ago. Many jawless fish died out at the end of the Devonian Period, almost 350 million years ago.

Birkenia

bir-KEEN-ee-a

- **When** 425 million years ago (Middle Silurian)
- **Fossil location** Europe
- **Habitat** Freshwater pools and streams
- **Length** 2 in (6 cm)

Although *Birkenia* did not have fins, it was still an active swimmer in the pools and streams in which it lived. It fed on the remains of dead plants and animals, probably sucking in scraps with its gaping mouth. Unlike many other jawless fish that had bony head-shields, *Birkenia*'s head was covered in small scales.



Sacabambaspis

SAC-a-bam-BASP-iss

- **When** 490 million years ago (Early Ordovician)
- **Fossil location** Bolivia
- **Habitat** Coastal waters
- **Length** 12 in (30 cm)

This fish had a broad head-shield and a body that narrowed to end in a small fin. With this shape, it probably swam very much like a tadpole, sucking in scraps of food through its ever-open mouth. *Sacabambaspis* had sense organs that helped it to feel movement in the water, allowing it to judge the distance to its prey—and avoid predators.

Zenaspis

zen-ASP-iss

- **When** 410 million years ago (Early Devonian)
- **Fossil location** Europe
- **Habitat** Shallow seas and river mouths
- **Length** 10 in (25 cm)



Zenaspis had a horseshoe-shaped head protected by an armored shield, while the rest of its fairly flat body was protected by scales. Its eyes were placed close together on top of its head (a perfect position for spotting predators for a bottom-dwelling fish). Like lots of jawless fish, *Zenaspis* did not have teeth. Instead, its mouth, located on the underside of the body, was lined with bony plates. It probably fed on small creatures found on the seafloor or in river mouths.

Cephalaspis

SEFF-a-LASP-iss

- **When** 410 million years ago (Early Devonian)
- **Fossil location** Europe
- **Habitat** Freshwater pools and streams
- **Length** 9 in (22 cm)

This small fish lived at the bottom of pools or streams. Perhaps it moved its broad head-shield from side to side, stirring up mud as it searched for hidden worms and other creatures. It may also have fed on the waste of other animals that lived in the water. Pairs of scaly flaps balanced its body, and a fin on its back prevented it from rolling over.

